

BSc (Hons) in Computer Science
BSc (Hons) in Computer Systems
Engineering
Year2

Lecture 04 Worksheet

SE2012 – Object Oriented Design and Analysis

Semester 1, 2026

Objectives:

1. Identify Classes and properties using Verb Noun Analysis
2. Identify correct relationships and draw class diagrams.
3. Draw class diagrams to real world scenarios

Section 1 : Verb Noun Analysis

Consider the following scenarios. Do the Noun Verb analysis to identify the classes in the description.

Question A.

An Online bookshop sells different things. Currently it has books, magazines and CDs. All these products have some common types of information such as price, title etc.. Also each product may have specific information such as author, number of pages, artists, type etc..) .

The registered customers can place orders for products. The ordered products are usually delivered together, but sometimes several shipments are needed when the store runs out of products for a while. An order may contain several products.

The products are delivered to personal customers and payments are done at the delivery. The business customers will get their orders directly, and they will be sent an invoice afterwards. Both customer types can make the payment also by using credit card. When registering, a customer is prompted to give contact and personal information needed for authentication, delivery of orders and invoicing.

The bookshop system must be able to add and remove customers, add new products and view/ update existing products, make or remove orders and view existing orders, view and generate reports on the products.

Do the Noun Verb analysis to identify the classes in the description.

- a) Identify all the Nouns which can be potential classes
- b) Identify the set of potential classes, show clearly how the nouns were rejected using the 5 rules.

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Question B.



- Identify the Objects in the above Picture
- Group these Objects into Classes

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Section 2 : Class Relationships

Work with another student to find answers together to these questions. Draw your solution on a piece of paper. If you have doubts check with your Tutors. The Tutors will discuss possible solutions with you during the Tutorial session.

- 1) HotelRoom, Booking, Guest
- 2) AcademicStaff, Lecturer, Professor, Student
- 3) Exhibition, ArtWork, OilPainting, WaterColourPainting, Sculpture, Photograph.
- 4) Prescription, Patient, Doctor, Appointment
- 5) Cinema Complex, Film Hall, Seat, Booking, Filmgoer

Section 3: Class Diagrams

Read the following description create a class diagram consisting of upto five classes that you have identified. Include properties and methods based on the description provided. Indicate the relationships clearly. Note that not all of the five relationships will be there in the solution.

Question A.

A new Cable TV company called VTV is providing over 100 entertainment TV channels to customers. Any person can become a customer by registering with VTV. The customer has to provide customer details and the package (channels he wishes to see) during the registration. The customer is provided with a unique customer number. The customer details are stored on the customer table and the package details selected by the customer are stored in the subscription table. Once a month the VTV sends the customer a detailed bill. This is produced using the data stored in the Subscription Table, Customer Table, and Channel details Table. The customer can login to VTV's website and perform several online tasks. The customer can add new channels to the subscription he has. He can remove channels from his subscription. In both cases the Subscription Table should be updated. In addition, the customer can provide a list of channels that he wants to get a detailed weekly TV schedule by email. These details are also stored in the Subscription Table. The customer can pay the monthly bills online using a credit

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card. He has to provide the credit card details and the amount for the system. The payment details are stored in the Payments Table. Every week VTV sends customers who have requested detailed TV schedules the details by email. TV schedules are stored in the Channel details Table.

Question B.

Food Delight is a famous restaurant situated in Colombo suburbs area. Both Locals and tourists both come to enjoy breakfast, lunch and dinner from this delicious restaurants. To give a better service for all the customers it has a new facility availability to place online orders, and the order will be delivered to the customers door step.

Customer can view the menu available online. Customer can select the food items he wishes and to place the order along with filling the delivery and contact details. When the customer places the order the manager can view that order. The manager is responsible for the timely delivery of the order. the manager will forward the order to the head chef in the kitchen.

If all the ingredients are available, the head chef will assign the order to junior chefs. If the ingredients are not available to fulfil the order, the head chef will reorder the quantities and an automatic email will be forwarded to the manger indicating that the particular order will be delayed 30-45 minutes extra. The Manager can call the customer and inform the delay of the order.

Once the order is prepared, the delivery team can print the order details and the head chef informs the manager of the completion of the order. The delivery team will deliver the order to the customers. Payments will be handled online when the customer orders. The customer can use credit or debit card to do the online transaction.